

Integration of reciprocal functions



1 Write an equation for $f(x)$, given that the derivative of $f(x)$ is $\frac{1}{x}$.

2 Find the primitive function of the following:

a $\frac{-2}{x}$

b $\frac{6}{x}$

c $\frac{1}{7x}$

d $\frac{1}{x+5}$

3 Find the following indefinite integrals:

a $\int \frac{4}{x} dx$

b $\int \frac{-7}{x} dx$

c $\int \frac{3}{4x} dx$

d $\int \frac{1}{x-4} dx$

e $\int \frac{2}{x+3} dx$

f $\int \frac{2}{2x+3} dx$

g $\int \frac{-2}{x-4} dx$

h $\int \frac{1}{2x+3} dx$

i $\int \frac{1}{5-x} dx$

j $\int \frac{2}{3x+5} dx$

k $\int \frac{2x}{x^2-5} dx$

l $\int \left(\frac{8}{x} + 5 \right) dx$

4 Find the exact value of the following definite integrals:

a $\int_3^9 \frac{3}{x} dx$

b $\int_4^6 \frac{1}{x-2} dx$

c $\int_3^5 \frac{2}{x+2} dx$

d $\int_{\frac{8}{3}}^{\frac{17}{3}} \frac{4}{3x-2} dx$

e $\int_1^3 \frac{5}{x+1} dx$

f $\int_3^5 \frac{x^2+1}{x} dx$

g $\int_0^3 \frac{1}{5+4x} dx$

h $\int_2^6 \left(2x + \frac{1}{x+2} \right) dx$

5 Determine the integral of the following functions:

a $f(x) = \frac{2x^2 + x - 3}{x}$, for $x > 0$.

b $f(x) = \frac{5x^3 + 2x^2}{x^3}$, for $x > 0$.

6 Consider the function $f(x) = x \ln x$.

a Differentiate $f(x) = x \ln x$.

b Hence, find $\int \ln x dx$.



Reverse chain rule

7 Consider the function $f(x) = \ln(x^2 - 4)$.

a Use the chain rule to differentiate $f(x)$.

b Hence, find $\int \frac{6x}{x^2 - 4} dx$.

8 Consider the function $f(x) = \ln(2x^2 + 4x)$.

a Use the chain rule to differentiate $f(x)$.

b Hence, find $\int \frac{x+1}{2x^2 + 4x} dx$.

9 Consider the function $f(x) = \ln(3x^3 + 5x^2)$.

a Use the chain rule to differentiate $f(x)$.

b Hence, find $\int \frac{18x^2 + 20x}{3x^3 + 5x^2} dx$.

10 Find the integral of the following functions:

a $\int \frac{4}{4x-3} dx$

b $\int \frac{15}{3x-2} dx$

c $\int \frac{-8}{2x+1} dx$

d $\int \frac{4x}{x^2+1} dx$

e $\int \frac{3x^2}{x^3+1} dx$

f $\int \frac{8x}{2x^2-3} dx$

g $\int \frac{2x^2}{2x^3+7} dx$

h $\int \frac{6x^2+3}{2x^3+3x} dx$

Area under the curve

11 Consider the function $y = \frac{2}{x}$.

a Sketch a graph of the function for $x > 0$.

b Find the exact area bounded by the curve, the x -axis, and the lines $x = 2$ and $x = 3$.

12 Consider the function $y = \frac{1}{x-2}$.

a Sketch a graph of the function for $x > 0$.

b Find the exact area bounded by the curve, the x -axis, and the lines $x = 3$ and $x = 5$.

13 Consider the function $y = \frac{1}{x+3}$.

a Sketch a graph of the function $y > 0$.

b Find the exact area bounded by the curve, the x -axis, and the lines $x = 2$ and $x = 4$.



14 Consider the function $y = \frac{1}{2x+5}$.

- a Sketch a graph of the function $y > 0$.
- b Find the exact area enclosed by the curve, the x -axis, and the lines $x = 3$ and $x = 4$.

15 Consider the function $y = \frac{1}{4-3x}$.

- a Sketch a graph of the function.
- b Find the exact area enclosed by the curve, the x -axis, and the lines $x = 3$ and $x = 5$.

16 Consider the function $y = \frac{2}{3x+5}$.

- a Sketch a graph of the function $x > 0$.
- b Find the exact area enclosed by the curve, the coordinate axes and the line $x = 4$.

17 Consider the function $y = x + \frac{1}{x}$.

- a State the domain of the function.
- b State the value the function approaches as $x \rightarrow \infty$.
- c Calculate the exact area enclosed by the function, the x -axis, and the lines $x = 2$ and $x = 12$.

18 Find the exact area enclosed by the function $f(x) = x + \frac{5}{x+5}$, the x -axis and the lines $x = 0$ and $x = 2$.

19 Consider the function $f(x) = \frac{6x^5 - 4x^2 + x}{2x^2}$, for $x > 0$.

- a Simplify and hence determine $\int f(x) dx$.
- b Find the exact area bounded by $f(x)$, the x -axis and the lines $x = 1$ and $x = 2$.

20 Consider the functions $y = x$ and $y = \frac{3}{x}$, for $x > 0$.

- a Sketch the graphs of these functions on the same coordinate axes.
- b At what x -value do the two functions intersect?
- c Find the area between the two functions, the x -axis and the line $x = 3$.

21 Consider the functions $y = -x^2$ and $y = \frac{1}{4-x}$ for $0 \leq x \leq 3$.

- Sketch the graphs of these functions on the same coordinate axes.
- Calculate the exact area bound by the curves between $x = 1$ and $x = 3$.

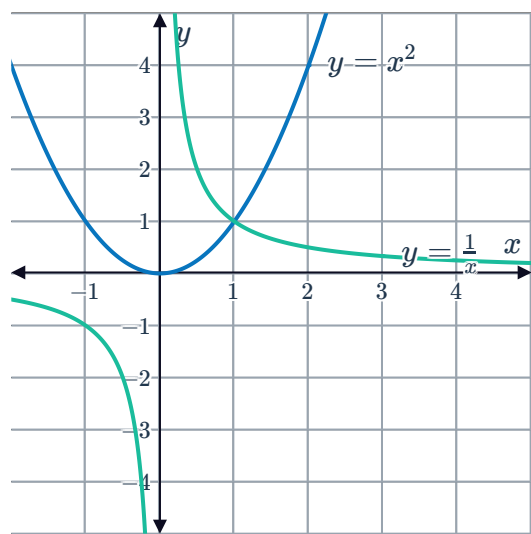
22 Consider the functions $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x}$.

- Sketch the graphs of the two functions on the same coordinate axes.
- At what x -value do the two functions intersect?
- Find the area of the region bounded by the two curves, the y -axis, and the line $y = 2$.

23 Consider the graph of the functions $y = x^2$ and $y = \frac{1}{x}$:

Solve for the exact value of k such that:

$$\int_1^k \frac{1}{x} dx = \int_0^1 x^2 dx$$



24 Consider the function $f(x) = \ln(x^2 + 1)$.

- Differentiate $f(x)$.
- Hence, find $\int \frac{x}{x^2 + 1} dx$.
- Find the area bounded by $g(x) = \frac{x}{(x^2 + 1)}$, the x -axis and the lines $x = 2$ and $x = 3$.

25 Consider the function $f(x) = \ln(\sin(x) + 1)$.

- Differentiate $f(x)$.
- Hence, find $\int \frac{\cos x}{\sin x + 1} dx$.
- Find the area bounded by $g(x) = \frac{\cos x}{\sin(x) + 1}$, the x -axis and the lines $x = 0$ and $x = \frac{\pi}{6}$.